KUNAL JAISWAL

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PROFESSIONAL SUMMARY

Software Engineer with 8+ years of production experience across backend and full-stack systems, distributed services, cloud automation, and AI/LLM applications. Led engineering initiatives spanning TypeScript/Node.js, React, Java, Azure, CI/CD, RAG, document processing, speech, and computer vision, with measurable improvements in productivity, reliability, and performance.

EXPERIENCE

Terawe Technologies | Lead Software Engineer — August 2022 – January 2026

- **Federated AI Platform Architecture:** Led AI solution development for law-enforcement and education use cases, designing a federated, multi-tenant platform across customer-deployed and centrally hosted Gov Cloud environments with tenant data isolation, secure authentication, and inter-instance communication.
- **RAG & Retrieval Systems:** Architected enterprise RAG pipelines using RAG-Fusion, HyDE, and parent-child chunking across multiple Qdrant vector databases to improve contextual retrieval and answer quality.
- **Multimodal AI & Document Intelligence:** Integrated LLaMA Vision for image-based queries and visual question answering; built document summarization, key-insight extraction, and knowledge-base navigation pipelines using OpenAI and Mistral models.
- **Real-Time Conversational AI:** Built an interactive AI sales simulation platform integrating Azure Speech Service, Silero VAD, WebSockets, and a fine-tuned Gemma model for context-aware voice conversations; also developed Whisper-based transcription, speaker diarization, and video-search workflows.
- **AI Evaluation & Full-Stack Delivery:** Developed an AI evaluation framework combining ground-truth metrics and LLM-as-a-Judge methods across classification, RAG, summarization, and entity extraction; deployed an LLM-based grading module that reduced manual grading effort by 70%, alongside Next.js dashboards and NestJS services on Azure App Service.

Finoa GmbH | Sr. Software Engineer — February 2022 – August 2022

- **High-Throughput Data Ingestion:** Built and deployed NestJS microservices ingesting real-time cryptocurrency prices from CoinGecko and other public APIs, storing millions of price records in a time-series PostgreSQL database.
- **Database & Throughput Optimization:** Improved ingestion throughput and reduced database load by over 40% through in-memory caching and batched database writes, reducing redundant operations and improving persistence efficiency.
- **Asynchronous Job Processing:** Implemented RabbitMQ-based job orchestration for background workloads, incorporating scheduling, retries, and failure handling to decouple processing tasks and improve resilience.
- **Backend Architecture & Observability:** Designed modular TypeScript services using dependency injection and validation pipelines; partnered with DevOps to establish CI/CD pipelines, Grafana-based

monitoring, centralized logging, and real-time alerting. Mentored junior engineers and participated in technical interviews.

Uberall GmbH | **Squad Lead Engineering** June 2021 – December 2021

- **Technical Leadership & Architecture:** Led a cross-functional team of five engineers building a local-business advertising platform and end-to-end “Near Me” discovery experiences, driving technical direction, architectural decisions, and feature delivery.
- **Serverless Backend Engineering:** Developed Node.js and TypeScript backend services using AWS Lambda, structuring application logic into maintainable components to support local-business discovery and advertising workflows.
- **Reusable Frontend Architecture:** Built responsive React and TypeScript applications using component-based architecture, reusable hooks, and state-management patterns to improve code reuse and maintainability.
- **Product Delivery & Engineering Leadership:** Partnered with product managers to prioritize features based on user feedback and business goals, streamlined onboarding and usability flows to support platform adoption, and mentored engineers while contributing to technical interviews.

Microsoft Corporation | **Sr. Software Engineer** April 2018 – April 2021

- **CI/CD Architecture & Technical Leadership:** Served as a subject-matter expert for CI/CD architecture and automation across multiple engineering teams, evolving deployment standards and DevOps practices across global teams.
- **Deployment Automation & Optimization:** Built a modular Groovy and PowerShell automation framework that reduced deployment time by 60%, standardizing delivery workflows across environments.
- **Idempotent API Design & Performance:** Replaced legacy update operations with an Admin API-based workflow that improved performance by 2×, ensured idempotency, simplified debugging, and was adopted across the organization.
- **Release Engineering & Deployment Safety:** Designed multi-stage Jenkins pipelines incorporating BVT, CTQ, and BCDR validations, rollback safeguards, and zero-touch deployment workflows to improve release consistency and reduce manual intervention.
- **Self-Healing Automation & Observability:** Engineered Jenkins bots to abort stale processes, recover failed jobs, and trigger alerts, reducing response time for critical failures by over 40%; built SQL Server data workflows and Power BI dashboards to automate operational logic and surface DevOps performance bottlenecks.

UHG – Optum | **Software Developer** February 2016 – April 2018

- **Backend Services & API Development:** Developed Node.js backend services and REST APIs integrating MongoDB and external REST services to support application data flows and system integrations.
- **Automated Testing & Quality Engineering:** Implemented backend and frontend testing using Mocha, Chai, Cucumber/Gherkin, Selenium, and Karma/Jasmine, applying TDD and ATDD practices to validate application behavior across the stack.
- **Custom Test Automation & CI/CD:** Built a JavaScript framework that converted Excel-driven test data into Gherkin scripts and integrated it with Jenkins and Selenium to automate UI validation; established Jenkins build and validation pipelines and developed AngularJS frontend components.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML, CSS

Backend & Frontend: Node.js, NestJS, Express, Spring Boot, Spring MVC, React, Next.js

AI / ML: LLMs, RAG, OpenAI APIs, Hugging Face, Transformers, Whisper, Vector Search, Speaker Diarization, Computer Vision

Cloud & Infrastructure: Azure, Azure DevOps, Kubernetes, Docker, Jenkins, OpenShift, CI/CD

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Cosmos DB, Qdrant, Neo4j

Testing: Jest, JUnit, Mockito, Mocha/Chai, Cucumber/Gherkin, Jasmine/Karma, Selenium, Puppeteer

EDUCATION

University of Texas at Dallas | M.S. in Computer Engineering

University at Buffalo, SUNY | B.S. in Computer Science

Honors: Dean's List (Spring 2013); Excellence Award (2020)